

ROTH, Germany

WMO: 107650

Lat: 49.2162N

Lon: 11.1035E

Elev: 1270

StdP: 14.03

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.4	14.0	5.5	7.7	9.9	10.8	10.0	16.7	22.5	43.9	19.6	39.5	2.8	130	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	88.1	66.7	84.2	65.2	80.8	64.1	68.4	82.8	67.0	80.3	65.5	77.5	5.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	94.1	71.2	62.6	89.0	70.5	61.1	84.2	69.7	33.4	83.5	32.2	80.1	31.0	77.6	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.3	12.1	1.1	93.7	6.0	3.6	-3.2	96.2	-6.7	98.3	-10.1	100.3	-14.5	102.9		
(5)			0.9	71.4	6.0	1.8	-3.4	72.7	-6.9	73.8	-10.3	74.8	-14.7	76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	31.2	33.5	40.0	47.9	56.2	62.2	65.4	64.8	56.6	48.6	39.3	32.1		
	DBStd	14.23	8.72	8.65	7.62	7.19	6.81	6.80	5.97	5.67	6.13	7.07	6.73	7.91		
	HDD50	2511	583	462	317	122	19	1	0	0	12	113	326	556		
	HDD65	6336	1048	882	774	513	282	134	70	72	258	510	772	1021		
	CDD50	1866	0	0	8	59	213	366	477	459	211	69	4	0		
	CDD65	216	0	0	0	0	11	49	81	67	7	1	0	0		
	CDH74	2915	0	0	0	22	228	661	1006	869	124	5	0	0		
	CDH80	912	0	0	0	2	45	195	357	296	17	0	0	0		
	WSAvg	5.0	5.4	5.6	5.9	5.0	5.0	4.8	4.8	4.4	4.3	4.3	4.8	5.1		
	PrecAvg	29.3	2.1	1.9	2.0	1.7	2.8	3.2	3.4	2.9	2.2	2.2	2.4	2.4		
(15) Precipitation	PrecMax	41.5	4.4	4.8	6.1	4.4	5.7	5.4	5.5	6.1	5.4	6.9	5.9	5.9		
	PrecMin	21.7	0.2	0.4	0.3	0.4	0.4	1.1	0.9	1.1	0.4	0.5	0.0	0.5		
	PrecStd	4.4	1.1	1.0	1.3	0.9	1.3	1.2	1.2	1.2	1.1	1.3	1.1	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	57.9	66.3	76.7	84.6	90.5	92.6	94.7	82.5	72.9	59.4	51.5		
		MCWB	47.0	47.9	51.4	56.4	63.9	67.5	68.0	67.5	64.6	60.4	52.6	46.8		
	2%	DB	49.8	52.4	61.2	72.2	80.0	85.7	88.2	86.7	77.7	67.5	55.2	48.7		
		MCWB	45.8	45.1	49.4	55.1	61.8	66.1	66.8	66.2	63.1	58.3	49.9	45.7		
	5%	DB	46.4	48.9	56.6	67.7	75.8	81.6	84.0	82.5	73.6	63.0	52.0	45.8		
		MCWB	42.9	43.9	47.5	53.7	59.7	64.3	65.5	64.8	60.9	55.7	48.4	43.2		
	10%	DB	43.0	45.5	52.2	63.0	71.4	77.6	79.6	78.8	69.6	59.5	49.1	42.7		
		MCWB	40.4	41.8	45.8	51.0	57.5	62.8	64.3	63.9	59.3	54.0	46.1	41.1		
	0.4%	WB	48.8	48.6	53.1	58.7	65.7	70.4	70.2	70.1	66.9	62.1	53.6	48.5		
		MCDB	52.0	55.2	61.4	72.5	80.1	82.9	85.7	84.9	77.2	70.7	58.7	50.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.9	46.5	50.7	56.2	63.2	67.9	68.4	68.0	64.2	59.1	50.9	45.9		
		MCDB	49.4	51.0	58.7	69.2	77.3	81.7	83.0	82.7	74.9	65.5	54.2	48.3		
	5%	WB	43.1	44.2	48.6	54.0	61.1	65.8	66.9	66.4	62.3	56.9	48.7	43.4		
		MCDB	45.8	48.2	55.1	65.5	73.2	78.3	79.9	79.5	71.2	62.3	51.6	45.3		
	10%	WB	40.4	42.0	46.3	52.0	58.9	63.8	65.4	64.7	60.1	54.7	46.3	40.8		
		MCDB	42.8	45.4	51.5	62.1	69.3	75.1	76.9	75.9	67.9	58.8	48.9	42.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	14.0	17.2	21.8	21.9	21.5	21.5	21.8	20.1	16.9	11.0	9.6		
		MCDBR	13.9	19.2	26.2	30.4	30.3	29.9	30.4	30.4	27.8	22.0	16.1	12.3		
		MCWBR	10.7	12.9	15.7	15.6	14.2	13.1	11.9	12.9	14.4	13.7	11.3	9.7		
		MCDBR	13.0	17.0	22.7	27.5	26.9	26.5	25.6	26.2	23.8	19.5	14.8	11.2		
	5% WB	MCWBR	10.5	12.0	14.1	14.7	13.2	12.2	10.9	11.8	12.7	12.5	10.9	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.326	0.361	0.393	0.401	0.404	0.411	0.405	0.377	0.353	0.327	0.300		
	taud		2.435	2.380	2.325	2.267	2.274	2.295	2.270	2.308	2.364	2.423	2.457	2.437		
	Ebn at Noon		240	260	268	270	272	272	268	264	259	246	228	225		
	Edh at Noon		20	27	33	39	41	40	40	37	32	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		281	546	898	1354	1594	1772	1680	1461	1059	635	315	223		
	RadStd		45	88	114	179	181	144	157	139	137	76	39	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.70	N/A	N/A	+90	+32

Nomenclature: See separate page